Seminar Report

Supplier Selection using AHP and ANP: A Review and an Improved Model

Review of "A supplier selection decision model using multi-criteria decision analysis in a small manufacturing company" (Rodrigues, Šírová and Mugurusi, 2022)

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CERTIFICATE

This is to certify that the seminar report entitled **"Supplier Selection using AHP and ANP: A Review and an Improved Model"** submitted by **Sujal Negi (Roll No: 123ME0023)** for the course **Seminar (Course Code: ME496)** at the Indian Institute of Information Technology, Design and Manufacturing, Kurnool, is a bonafide record of the work carried out by him during the Academic Year 2026–2027.

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ABSTRACT

Choosing a supplier for a critical spare part is a decision with several goals that pull against each other: low price, short delivery time, good quality and reliable service. The paper reviewed in this seminar, Rodrigues, Šírová and Mugurusi (2022), studies this decision in a small Czech car-parts manufacturer that must buy copper rods from one of three suppliers. Three managers compared six attributes, grouped into a Logistic-Economic criterion and a Quality-Sustainability criterion, using Saaty's 1–9 pairwise scale. The authors solved the problem with the Analytic Hierarchy Process (AHP), which recommended Supplier 2, and with the Analytic Network Process (ANP), which recommended Supplier 1. They chose the ANP answer.

In this work every score in the paper was recomputed from its own published tables. The AHP result was reproduced to four decimal places. The ANP result was also reproduced exactly, but only by giving Quality-Sustainability five times the weight of Logistic-Economic inside an ordinary AHP calculation. This shows that the change of winner comes from one group-weight judgment, not from the network structure of ANP. The decision has a single tipping point: Supplier 1 wins when Logistic-Economic receives less than 44.2% of the weight, and Supplier 2 wins above it. The re-analysis also found that measurable quantities (price, lead time and defect rate) were turned into opinions, which made a 1.5 times price difference look like a 7 times difference, and that four of the eight comparison tables exceed Saaty's 5% consistency limit for three items.

An improved model is proposed and built as an offline web demo. It scores price, lead time and defect rate from real numbers, runs a Monte Carlo robustness test of 5,000 random trials, draws a priority map for any mix of cost, quality and speed, and simulates the cost of each buying plan over 24 months. The results show that Supplier 3 never wins, that Supplier 1 and Supplier 2 are close (Supplier 2 wins 58% of trials with the paper's opinions and 54% with real numbers), and that over two years Supplier 2 is the cheaper plan unless a single bad rod costs the company more than about €5,250. The improved model therefore gives managers a confidence level and a cost view, not just one ranking.

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Chapter 1

Introduction

1.1 Supplier Selection in Manufacturing

In manufacturing, purchasing is not only about the lowest price. A spare part that arrives late can stop a production line, and a poor-quality part can damage machines or reach the customer. The buying decision has to balance cost, delivery time, quality, service and sustainability at once. Such problems are called *multi-criteria decision making* (MCDM) problems.

1.2 The Case Company

The paper reviewed here studies a small industrial company in the Czech Republic with about 400 employees that makes components for the automotive industry. Its maintenance department buys and stores spare parts, and safety stock levels are set from the previous year's demand. The company faced high costs and long lead times for spare parts. Its most critical part, chosen after an ABC analysis, is the **copper rod**. If a rod is missing, production stops.

Three suppliers can deliver the rods: Supplier 1 from Germany, Supplier 2 from the Central Bohemian Region and Supplier 3 from the North Bohemian Region of the Czech Republic. According to the paper, Supplier 1 delivers the best quality but has the highest price and is the farthest away. Unit prices range from €230 to €350, the yearly spend on rods was about €25,300, delivery times are steady between 3 and 7 days, and the share of low-quality parts rose at all three suppliers from 5–13.33% in 2017 to 6.67–15% in 2018.

1.3 Why AHP and ANP

AHP and ANP, both developed by Saaty, turn a hard decision into many small questions of the form "how much more important is A than B?", which managers can answer from experience, and combine the answers into a score for each option. The paper applies both methods and compares their rankings.

1.4 Aim of This Seminar

The aim of this seminar is to understand what the paper did, check its results by rebuilding the model from the published tables, identify its limitations, and propose and demonstrate an improved model that gives the company a more reliable and more useful answer.

Chapter 2

Literature Review

2.1 Analytic Hierarchy Process (AHP)

Saaty (1980) introduced AHP as a way to structure a decision as a tree: the goal at the top, criteria and sub-criteria in the middle, and the alternatives at the bottom. At every level, the decision maker compares items in pairs on a scale from 1 (equally important) to 9 (extremely more important). The answers form a *pairwise comparison matrix* A, in which $a_{ji} = 1/a_{ij}$. The weights w of the items are the principal eigenvector of A:

$$A\,w = \lambda_{\max}\,w. \tag{2.1}$$

Because people are not perfectly consistent, AHP also measures consistency. The consistency index and consistency ratio are

$$CI = \frac{\lambda_{\max} - n}{n - 1}, \qquad CR = \frac{CI}{RI}, \tag{2.2}$$

where n is the size of the matrix and RI is a random index. Saaty recommends $CR \le 0.05$ for 3×3 matrices, 0.08 for 4×4 and 0.10 for larger matrices. Many software tools accept up to 0.10 in all cases.

2.2 Analytic Network Process (ANP)

Saaty (2005) extended AHP to the Analytic Network Process. ANP drops the strict tree and allows criteria to influence each other and to receive feedback from the alternatives. Items are grouped into *clusters*, and the influences are stored in a *supermatrix* that is raised to powers until it converges. ANP is useful when criteria really depend on each other, for example when a supplier's distance affects both its delivery time and its availability.

2.3 MCDM for Spare Parts and Suppliers

Gajpal, Ganesh and Rajendran (1994) and Molenaers et al. (2012) used AHP to rank spare parts by criticality. Kabir and Hasin (2011) showed that fuzzy AHP can capture the vagueness of human judgments, and Rezaei, Fahim and Tavasszy (2014) used fuzzy AHP to select airline suppliers. Koksalmis and Kabak (2019) reviewed objective ways to weight decision makers in a group.

2.4 Supplier Selection and Order Allocation

Alegoz and Yapicioglu (2019) argue that an effective supplier selection approach should not only pick the right supplier but also decide how much to order from each, considering quantity discounts and fast-service options. This view is important for the present work, because when two suppliers are close, splitting the order can be safer than choosing only one.

2.5 Summary of the Reviewed Literature

Table 2.1 summarises the reviewed works and what each leaves open for the present case.

Table 2.1: Summary of reviewed literature

Study	**Main idea**	**What it leaves open here**
Saaty (1980)	AHP: tree, 1–9 pairwise scale, eigenvector weights, consistency ratio	One fixed answer; no measure of how sure it is
Saaty (2005)	ANP: network, clusters, feedback, supermatrix	Needs many extra judgments; easy to apply without real interdependence
Kabir & Hasin (2011)	Fuzzy AHP vs AHP for inventory classes	Not applied to this supplier case
Alegoz & Yapicioglu (2019)	Supplier selection with order allocation	No pairwise judgments of managers
Rodrigues et al. (2022)	AHP vs ANP for a copper rod supplier in a small company	No sensitivity, no use of measured data, no cost view over time

Source: compiled by the author from the cited works

Chapter 3

The Paper's Model

3.1 Decision Makers

Three managers took part, because the purchase affects all three departments:

- **Maintenance manager** (4 years in role, senior electrical engineer). Leads the purchase, negotiates prices and delivery terms and controls spare-part stock.
- **Production manager** (5 years, senior mechanical engineer). Believes that line stops cost more than any delivery charge.

- **Quality manager** (10 years, senior mechanical engineer). Follows a zero-defect policy and rates part quality far above price and lead time.

Their individual judgments were combined by the geometric mean into one group judgment for every comparison.

3.2 Criteria and Attributes

Six attributes were selected from the literature and grouped into two hybrid criteria (clusters), as shown in Figure 3.1:

- **C1 Logistic-Economic:** delivery cycle time, supplier availability (response to emergency orders) and unit price.
- **C2 Quality-Sustainability:** part quality (share of parts accepted at inspection), aftersales customer service and sustainability (remanufacture rate of scrapped parts).

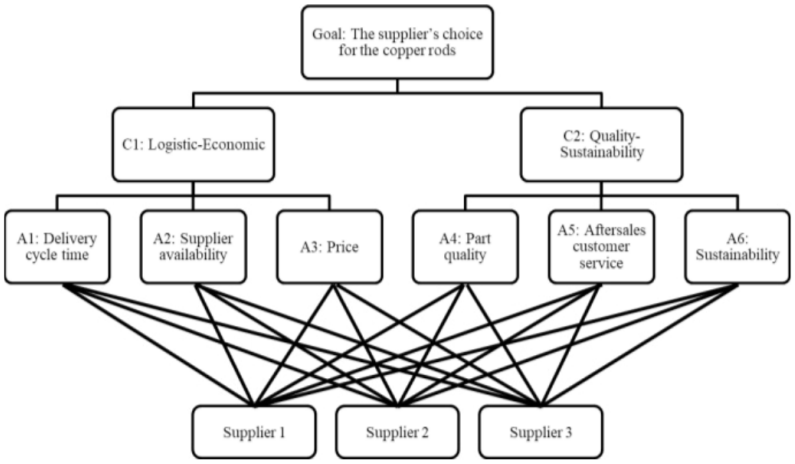


Figure 3.1: Hierarchical tree of the AHP model (goal, two criteria, six attributes, three suppliers)

Source: Rodrigues, Šírová and Mugurusi (2022), Figure 1, p. 2775

3.3 Pairwise Judgments

The paper publishes the group judgments as eight 3×3 matrices (its Tables 2–9). Table 3.1 lists the upper half of each matrix (the lower half is the reciprocal) together with the priority vector recomputed in this work.

3.4 AHP and ANP Models

In the AHP model the two criteria are treated as equally important. For ANP, the authors describe a network in which the criteria clusters are linked and influence each

Table 3.1: Group judgments from the paper and the recomputed priorities

Matrix (items 1, 2, 3)	a_{12}	a_{13}	a_{23}	**Priorities**
C1: Delivery, Price, Availability	5	4	1/3	0.674, 0.101, 0.226
C2: Aftersales, Quality, Sustain.	1/3	2	3	0.249, 0.594, 0.157
Delivery time: S1, S2, S3	1/5	1/3	3	0.105, 0.637, 0.258
Part quality: S1, S2, S3	3	7	5	0.649, 0.279, 0.072
Availability: S1, S2, S3	1/3	1/3	1	0.143, 0.429, 0.429
Aftersales: S1, S2, S3	3	3	2	0.594, 0.249, 0.157
Price: S1, S2, S3	1/6	1/8	1	0.067, 0.444, 0.489
Sustainability: S1, S2, S3	3	5	3	0.637, 0.258, 0.105

Source: judgments from Rodrigues et al. (2022), Tables 2–9; priorities recomputed by the author (Excel sheet T1)

other, with feedback from the alternatives, so that C1 and C2 are no longer equally important. All calculations were done in the SuperDecisions software. The network is shown in Figure 3.2.

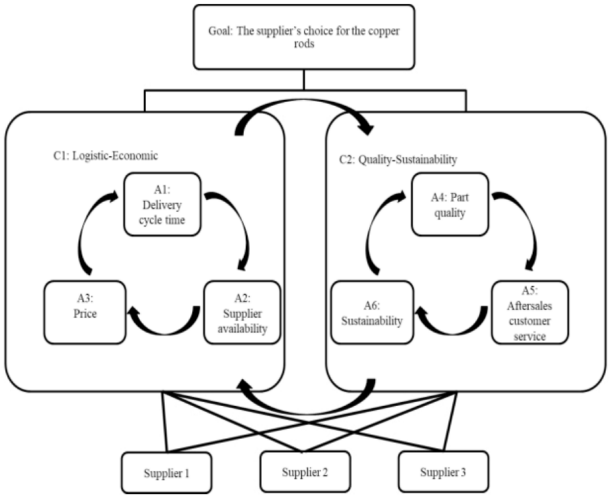


Figure 3.2: ANP model: the two clusters influence each other and receive feedback from the suppliers

Source: Rodrigues, Šírová and Mugurusi (2022), p. 2776 (printed as "Figure 1" on that page; called Figure 2 in the paper's text)

3.5 Results Reported in the Paper

Table 3.2 shows the final rankings printed in the paper. AHP recommends Supplier 2 and ANP recommends Supplier 1. The authors recommended the ANP result because ANP is more suitable for non-hierarchical relations.

The authors also note two difficulties: managers were reluctant to apply the methods to their confidential data, and keeping the judgments consistent was hard because they

Table 3.2: Final scores reported by the paper (Tables 10–12)

Method	Supplier 1	Supplier 2	Supplier 3	Winner
AHP	0.371467	0.419439	0.209094	Supplier 2
ANP	0.546076	0.318691	0.135233	Supplier 1

Source: Rodrigues et al. (2022), Tables 10–12

were using the methods for the first time.

Chapter 4

Re-analysis of the Paper

4.1 Method of Re-analysis

All eight matrices of Table 3.1 were entered into a program written for this seminar. Priorities were computed with the power method (repeated multiplication of the matrix with a vector until it stops changing), which gives the principal eigenvector. The consistency ratio was computed with $RI = 0.52$ for $n = 3$, the value used by SuperDecisions. With this value the program reproduces every inconsistency percentage printed in the paper (8.25%, 5.16%, 3.70%, 6.24%, 0.00%, 5.16%, 0.89% and 3.70%).

The final score of supplier i is

$$S_i(w) = w \sum_{a \in C1} v_a \, p_{a,i} \; + \; (1 - w) \sum_{a \in C2} v_a \, p_{a,i}, \tag{4.1}$$

where w is the weight of C1, v_a is the weight of attribute a inside its group, and $p_{a,i}$ is the score of supplier i on attribute a.

4.2 AHP Result Reproduced

With $w = 0.5$, Equation 4.1 gives 0.3715, 0.4194 and 0.2091, which is exactly the paper's AHP result (Table 4.1).

4.3 The ANP Result Is a Re-weighted AHP

Equation 4.1 is a straight line in w. Writing $S_i(w) = S_i(0) + w\,[S_i(1) - S_i(0)]$, the values at the two ends are

$$S(0) = (0.6334,\ 0.2683,\ 0.0983), \qquad S(1) = (0.1096,\ 0.5706,\ 0.3199).$$

Solving $S_1(w) = 0.546076$ gives $w = 0.1667 = 1/6$. At this value the other two scores are 0.3187 and 0.1352, which also match the paper. So the published ANP ranking is **exactly** an AHP ranking in which Quality-Sustainability is five times as important as Logistic-Economic (1 : 5). No influence between attributes or feedback from the alternatives is needed to produce it.

Table 4.1: Paper results compared with the re-analysis

Calculation	S1	S2	S3	Winner
Paper, AHP	0.3715	0.4194	0.2091	Supplier 2
This work, $w = 1/2$	0.3715	0.4194	0.2091	Supplier 2
Paper, ANP	0.5461	0.3187	0.1352	Supplier 1
This work, $w = 1/6$	0.5461	0.3187	0.1352	Supplier 1

Source: paper values from Tables 10–11; rebuilt values: own calculation (Excel sheet T2)

This is the main finding of the re-analysis. The paper presents the change of winner as the effect of a deeper, network view of the problem. In fact it is the effect of one judgment about the relative importance of the two groups. Whether quality really is five times as important as logistics is exactly the question the managers disagree on, and the paper does not show that the answer depends on it.

4.4 The Tipping Point

Setting $S_1(w) = S_2(w)$ gives the weight at which the winner changes:

$$w^* = \frac{S_1(0) - S_2(0)}{[S_1(0) - S_2(0)] - [S_1(1) - S_2(1)]} = \frac{0.3651}{0.3651 + 0.4610} = 0.442. \tag{4.2}$$

Supplier 1 wins when Logistic-Economic receives less than 44.2% of the weight, and Supplier 2 wins above it (Table 4.2 and Figure 4.1). Supplier 3 is never first for any w. The paper's AHP point ($w = 0.5$) is only 0.058 away from the tipping point, and the lead of Supplier 2 there is just 0.048.

4.5 One-attribute Sensitivity

Each attribute's global weight was then changed alone, with the others scaled so that the total stays 1, to find how far it must move before the winner changes (Table 4.3). Delivery time, part quality, aftersales and sustainability can each flip the decision with a change of only 0.07–0.11, which confirms that the AHP answer is fragile.

4.6 Consistency of the Judgments

Table 4.4 and Figure 4.2 show the consistency ratio of all eight matrices. All are below the common 10% limit, but four (Tables 2, 3, 5 and 7) are above the 5% limit that

Table 4.2: Final score of each supplier for different weights of Logistic-Economic

Weight w	S1	S2	S3	Winner
0.0	0.633	0.268	0.098	Supplier 1
0.1	0.581	0.299	0.120	Supplier 1
0.2	0.529	0.329	0.143	Supplier 1
0.3	0.476	0.359	0.165	Supplier 1
0.4	0.424	0.389	0.187	Supplier 1
0.5	0.371	0.419	0.209	Supplier 2
0.6	0.319	0.450	0.231	Supplier 2
0.7	0.267	0.480	0.253	Supplier 2
0.8	0.214	0.510	0.276	Supplier 2
0.9	0.162	0.540	0.298	Supplier 2
1.0	0.110	0.571	0.320	Supplier 2

Source: own calculation with Equation 4.1 (Excel sheet T3, 21 points in steps of 0.05; demo, Sensitivity tab)

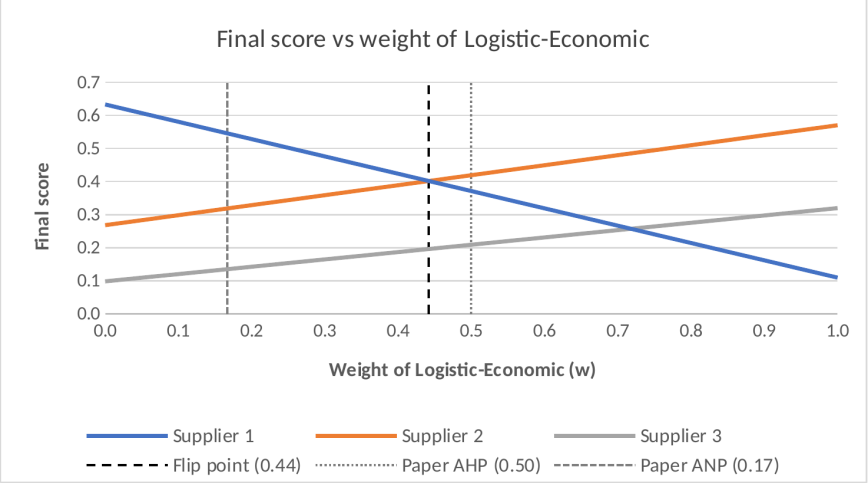


Figure 4.1: Final scores as the weight of Logistic-Economic changes. The winner flips at 0.44

Source: Excel chart of Table 4.2 (workbook sheet T3)

Table 4.3: One-attribute sensitivity at the paper's AHP setting ($w = 0.5$)

Attribute	Weight now	Flips at	Change	New winner
Delivery time	0.337	0.271	−0.066	Supplier 1
Price	0.050	0.834	+0.783	Supplier 3
Availability	0.113	never	–	–
Aftersales service	0.125	0.232	+0.107	Supplier 1
Part quality	0.297	0.378	+0.081	Supplier 1
Sustainability	0.079	0.182	+0.104	Supplier 1

Source: own calculation (Excel sheet T3; demo, Sensitivity tab)

Saaty recommends for three items. The comparison of C1 attributes (8.25%) is the least consistent, and it is the matrix that sets the weights of delivery time and price.

Table 4.4: Consistency ratio of each judgment matrix

Matrix	Paper	This work	Saaty limit (5%)
Table 2 (C1 attributes)	8.25%	8.25%	above 5%
Table 3 (C2 attributes)	5.16%	5.16%	above 5%
Table 4 (delivery time)	3.70%	3.70%	within 5%
Table 5 (part quality)	6.24%	6.24%	above 5%
Table 6 (availability)	0.00%	0.00%	within 5%
Table 7 (aftersales)	5.16%	5.16%	above 5%
Table 8 (price)	0.89%	0.88%	within 5%
Table 9 (sustainability)	3.70%	3.70%	within 5%

Source: paper values: Rodrigues et al. (2022), Tables 2–9; this work: recomputed with RI = 0.52 (Excel sheet T4)

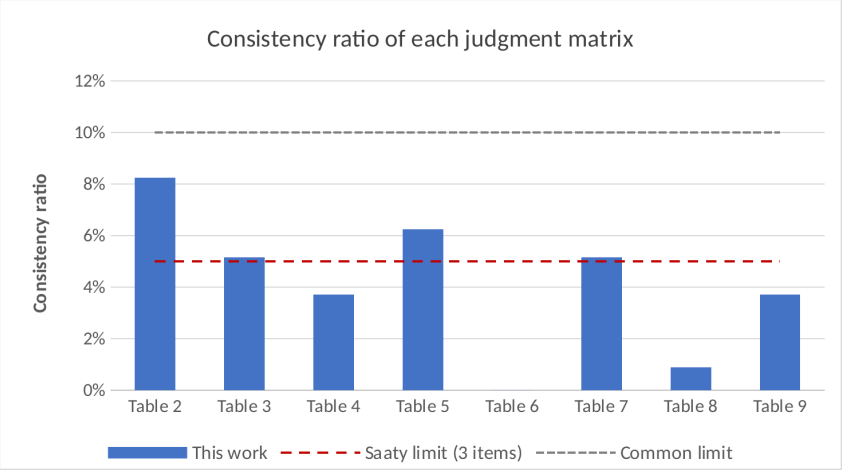


Figure 4.2: Consistency ratio of the eight judgment matrices

Source: Excel chart of Table 4.4 (workbook sheet T4)

4.7 Opinions in Place of Measurements

Price, delivery time and defect rate are measured by the company, yet the paper scores them with opinions. On the 1–9 scale, Supplier 1 was judged 6 and 8 times worse than Suppliers 2 and 3 on price, which gives it a price score of 0.067 against 0.489. But the real price ratio is at most 350/230 = 1.52. The verbal scale stretched a 1.5 times difference into a 7 times difference. The same effect appears for delivery time and quality (Figure 5.2, Section 5.2).

4.8 Summary of Research Gaps

Table 4.5: Research gaps found and the approach taken in this work

Gap	What was found	Approach in this work
ANP gives a different winner	ANP = AHP with groups $1:5$	Show the exact effect of the group weight
No sensitivity analysis	Winner flips at $w = 0.44$; several attributes can flip it	What-if slider and sensitivity table
Measured data used as opinions	$1.5\times$ price gap scored as $7\times$	Score price, lead time and defects from numbers
Inconsistent judgments	4 of 8 matrices above 5%	Live consistency check
One fixed ranking	Judgments are uncertain	Monte Carlo robustness test
No cost view over time	Ranking ignores line stops and bad parts	Two-year cost simulation
Equal weight for managers	4, 5 and 10 years of experience treated alike	Listed as future work

Source: own analysis of Rodrigues et al. (2022)

Chapter 5

Proposed Improvement

5.1 Objectives

1. **Rebuild and check** the paper's model from its own tables, including consistency.
2. **Use real numbers** for attributes that are measured: price, delivery time and defect rate.
3. **Measure how sure** the answer is: tipping point, sensitivity and a Monte Carlo robustness test.
4. **Show the long-term effect** of each buying plan on the company's costs over 24 months, including a split order.
5. Deliver all of this as a **simple tool** that managers can use without special software.

Figure 5.1 shows the flow of the proposed model.

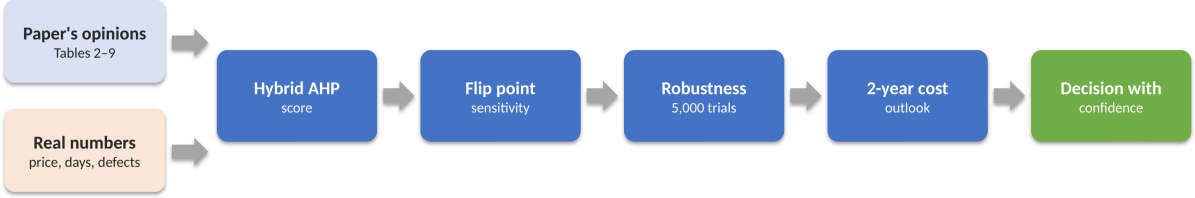


Figure 5.1: Flow of the proposed improved model

Source: own work

5.2 Hybrid Scoring

For attributes where lower values are better and real numbers x_i are available, the score of supplier i is

$$p_i = \frac{1/x_i}{\sum_k 1/x_k}. \tag{5.1}$$

This keeps the real ratios: a supplier that is 1.5 times dearer gets a price score 1.5 times lower. The other attributes (availability, aftersales and sustainability) keep the managers' judgments. The paper gives only ranges, so the values for each supplier were placed inside those ranges following the paper's own rankings (Table 5.1). The demo lets the user replace them with the company's real figures.

Table 5.1: Measured values used in the hybrid model

Measure	S1	S2	S3	Paper's range
Unit price (€)	350	230	235	230–350
Delivery time (days)	7	3	5	3–7
Bad parts 2017 (%)	5.00	8.33	13.33	5–13.33
Bad parts 2018 (%)	6.67	10.00	15.00	6.67–15

Source: ranges from Rodrigues et al. (2022), Section 3; per-supplier values estimated by the author inside those ranges (Excel sheet T5)

Table 5.2: Attribute scores from opinions and from real numbers

	Opinions (paper)			Real numbers		
Attribute	S1	S2	S3	S1	S2	S3
Price	0.067	0.444	0.489	0.249	0.379	0.371
Delivery time	0.105	0.637	0.258	0.211	0.493	0.296
Part quality	0.649	0.279	0.072	0.474	0.316	0.211

Source: opinions: recomputed from Rodrigues et al. (2022), Tables 4, 5 and 8; real numbers: Table 5.1 with $p_i = (1/x_i)/\sum_k 1/x_k$ (Excel sheet T5)

5.3 Robustness Test (Monte Carlo)

A judgment of "5" could easily have been a 4 or a 6, and nobody knows the true group weight. In each of 5,000 trials, every judgment is moved at random by -1, 0 or $+1$ step on the Saaty scale, the weight w of C1 is drawn at random between 0.1 and 0.9, every measured value is varied by up to $\pm 5\%$ (hybrid mode), and the winner is recorded. The output is the share of trials that each supplier wins, and the share in which it finishes second or third.

5.4 Company Priorities

To answer the question "if two suppliers are close, what should decide it?", the company's priorities are expressed as three numbers from 0 to 10: cost, quality and speed. Cost is

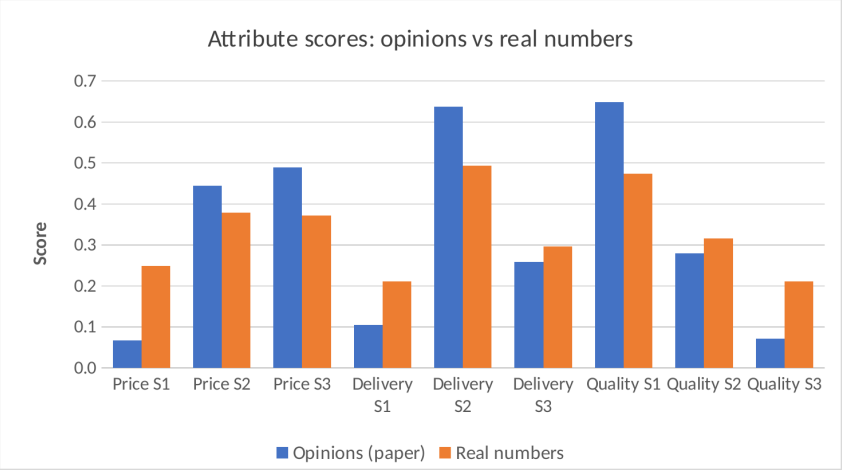


Figure 5.2: Scores on price, delivery time and part quality from opinions and from real numbers

Source: Excel chart of Table 5.2 (workbook sheet T5)

the price attribute, speed is delivery time plus availability, and quality is part quality, aftersales and sustainability (split inside each group by the paper's own weights). The three numbers are scaled to add up to 1 and used as attribute weights, and the suppliers are ranked again.

5.5 Two-year Cost Simulation

The ranking alone does not tell the company what each choice will cost. A month-by-month simulation over 24 months was therefore added. In every month t:

$$\text{rods needed} \sim \text{Poisson}(R/12), \tag{5.2}$$

$$\text{bad rods from supplier } i \sim \text{Binomial}(\text{rods}_i,\ d_i(t)), \quad d_i(t) = d_{i,2018} + g\,t/12, \tag{5.3}$$

$$\text{urgent needs} \sim \text{Poisson}(U/12). \tag{5.4}$$

Each bad rod or urgent need is a "wait". With probability q the shelf is empty, and the line then stops for the supplier's lead time L_i days, losing h hours per day at a cost of c per hour. The total cost is

$$\text{Cost} = \underbrace{\sum \text{rods} \times P_i(t)}_{\text{buying}} + \underbrace{\text{bad rods} \times c_b}_{\text{handling}} + \underbrace{\text{stop hours} \times c}_{\text{line stops}}, \tag{5.5}$$

where the price $P_i(t)$ rises by r per year to reflect the copper market. Each plan (only S1, only S2, only S3, or a split between S1 and S2) is simulated 400 times, and the mean and the 10th–90th percentile band are reported. Table 5.3 lists the inputs and where they come from.

Table 5.3: Inputs of the two-year simulation

Input	Value	Source
Rods needed per year, R	90	Paper: about €25,400 a year / average price
Rise in bad parts, g	1.67 pts/yr	Paper: 2017 to 2018 rise at every supplier
Copper price rise, r	5%/yr	Market: LME copper averaged about 32% more in 2026 than in 2025
Handling cost per bad rod, c_b	€150	Estimate (inspection, return, paperwork)
Urgent needs per year, U	4	Estimate
Chance shelf is empty, q	5%	Estimate
Hours lost per waiting day, h	8	Estimate (one shift)
Cost of one stop hour, c	€300	Estimate for a small plant

Source: as stated in the third column (Excel sheet T8)

Chapter 6

Demo and Results

6.1 Implementation

The improved model was built as a web application called **Supplier Decision Lab**. It is written in HTML, CSS and JavaScript, draws its charts with Chart.js 4.4.1 (bundled), and runs fully offline. All numbers live in one data file with their source. The values produced by the demo were exported to a Microsoft Excel workbook (`Seminar_Data_and_Charts.xlsx`), in which every table and chart of this report is kept with its source (sheets T1–T8). A small Node.js script with no dependencies serves the page, so the demo starts with one command in the project folder: `node server.js`, then open `http://localhost:8080`.

The demo has five tabs. The first shows what the paper has: the suppliers, the paper's result next to the rebuilt result, the decision tree and all eight judgment tables as sliders with a live consistency check. The other four show what is new: Sensitivity, Real numbers, Robustness and 2-year outlook. Every slider and input updates all charts at once.

6.2 Results: Real Numbers

Table 6.1 compares the paper's opinion-based scores with the hybrid scores. With real numbers the gap between Supplier 1 and Supplier 2 at equal group weights shrinks from 0.048 to 0.014, and the tipping point moves from 0.442 to 0.472. The overall picture does not change: Supplier 3 is last, and Suppliers 1 and 2 are close.

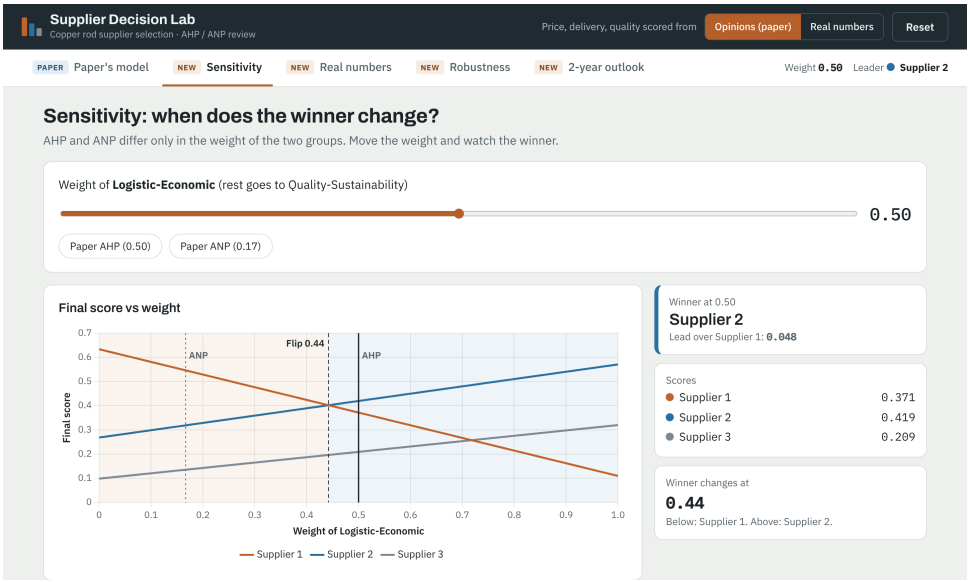


Figure 6.1: Developed demo, Sensitivity tab: weight slider, score lines, flip point and one-attribute table

Source: screenshot of the developed demo (own work)

Table 6.1: Final scores with opinions only and with real numbers

Model	S1	S2	S3	Flip point
Opinions, $w = 0.5$	0.371	0.419	0.209	0.442
Real numbers, $w = 0.5$	0.364	0.379	0.257	0.472
Opinions, $w = 1/6$	0.546	0.319	0.135	–
Real numbers, $w = 1/6$	0.474	0.320	0.206	–

Source: own calculation (Excel sheet T5; demo, Real numbers tab)

6.3 Results: Robustness

Table 6.2 and Figure 6.2 give the results of 5,000 random trials. Supplier 3 never wins and finishes last in about 76% of trials. Supplier 2 wins 58% of trials with the paper's opinions and 54% with real numbers, so neither of the two leading suppliers is a reliable winner (a share of at least 80% was used as the bar). Table 6.3 and Figure 6.3 show that the score difference between Supplier 1 and Supplier 2 is spread widely on both sides of zero (mean −0.05, standard deviation 0.19).

Table 6.2: Share of 5,000 trials in which each supplier finishes 1st, 2nd and 3rd

	Paper's opinions			**Real numbers**		
Supplier	1st	2nd	3rd	1st	2nd	3rd
Supplier 1	41.9%	34.2%	23.9%	46.0%	31.0%	23.0%
Supplier 2	58.1%	41.9%	0.0%	54.0%	46.0%	0.0%
Supplier 3	0.0%	23.9%	76.1%	0.0%	23.0%	77.0%

Source: demo, Robustness tab: 5,000 trials, seed 7 (Excel sheet T6)

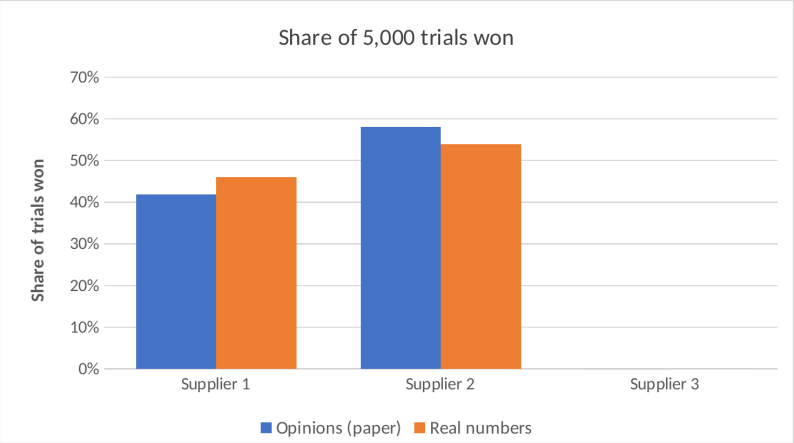


Figure 6.2: Share of 5,000 trials won by each supplier

Source: Excel chart of the "1st" columns of Table 6.2 (workbook sheet T6)

The test also shows what drives the result. If the weight of C1 is known to lie between 0.1 and 0.3 (quality first), Supplier 1 wins every trial. Between 0.6 and 0.9 (logistics first), Supplier 2 wins every trial. Between 0.3 and 0.7, Supplier 2 wins 66% with opinions and 57% with real numbers. The decision therefore depends almost entirely on how the company ranks quality against logistics, which is a management choice, not something the method can settle.

6.4 Results: Priorities

With the paper's opinions, a balanced mix (cost, quality and speed equal) ranks Supplier 2 first (0.432), Supplier 3 second (0.296) and Supplier 1 third (0.272). Cost-first and speed-first mixes also choose Supplier 2. Only a quality-first mix (2 : 8 : 2) chooses Supplier 1, with 0.453 against 0.350. Table 6.4 lists the four preset mixes.

Table 6.3: Number of trials by score difference (Supplier 1 minus Supplier 2)

S1 − S2 range	Trials (opinions)	Trials (real numbers)
−0.5 to −0.4	36	0
−0.4 to −0.3	519	0
−0.3 to −0.2	802	236
−0.2 to −0.1	782	1224
−0.1 to −0.0	768	1240
−0.0 to +0.1	768	1246
+0.1 to +0.2	777	1001
+0.2 to +0.3	500	53
+0.3 to +0.4	48	0
+0.4 to +0.5	0	0

Source: demo, Robustness tab, 5,000 trials, seed 7 (Excel sheet T6)

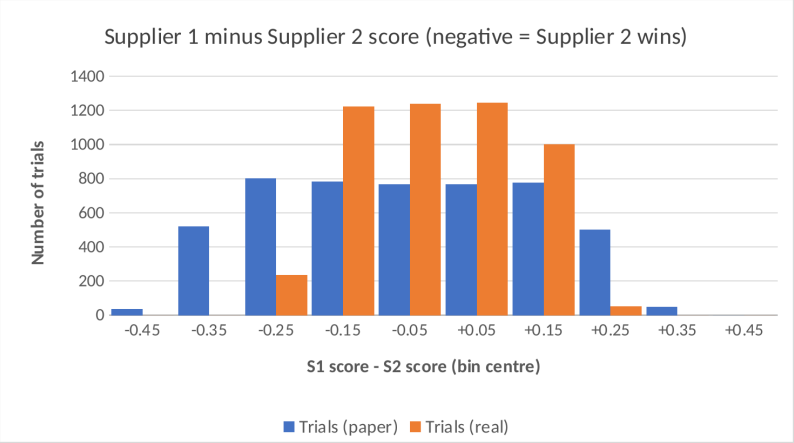


Figure 6.3: Score difference S1 minus S2 over 5,000 trials

Source: Excel chart of Table 6.3 (workbook sheet T6)

Table 6.4: Supplier scores for four company priorities

Priority	Cost : Quality : Speed	S1	S2	S3	Winner
Balanced	4 : 4 : 4	0.272	0.432	0.296	Supplier 2
Cost first	8 : 2 : 3	0.165	0.449	0.385	Supplier 2
Quality first	2 : 8 : 2	0.453	0.350	0.197	Supplier 1
Speed first	2 : 3 : 8	0.227	0.490	0.283	Supplier 2

Source: demo, 2-year outlook tab, company priority presets (opinion scores)

6.5 Results: Two-year Outlook

Tables 6.5 and 6.6 and Figures 6.4 and 6.5 give the 24-month results. Buying only from Supplier 2 is the cheapest plan at about €56,100. Buying only from Supplier 1 costs about €86,400: it has the fewest bad rods (14.8 against 21.0), but its higher price and longer delivery from Germany add both purchase cost and line-stop hours. A 50/50 split costs about €72,400, close to buying only from Supplier 3, but with far fewer bad rods.

Table 6.5: Cumulative cost of each buying plan (€, mean of 400 simulations)

Month	Only S1	Only S2	Only S3	Split 50/50
3	10,005	6,726	8,661	8,749
6	20,868	13,539	17,518	17,571
9	31,496	20,427	26,741	26,251
12	42,191	27,453	35,590	35,257
15	53,274	34,651	44,890	44,537
18	64,389	41,727	54,248	53,781
21	75,402	49,028	63,011	62,895
24	86,426	56,120	72,321	72,361

Source: demo, 2-year outlook tab, inputs of Table 5.3 (Excel sheet T7 lists all 24 months)

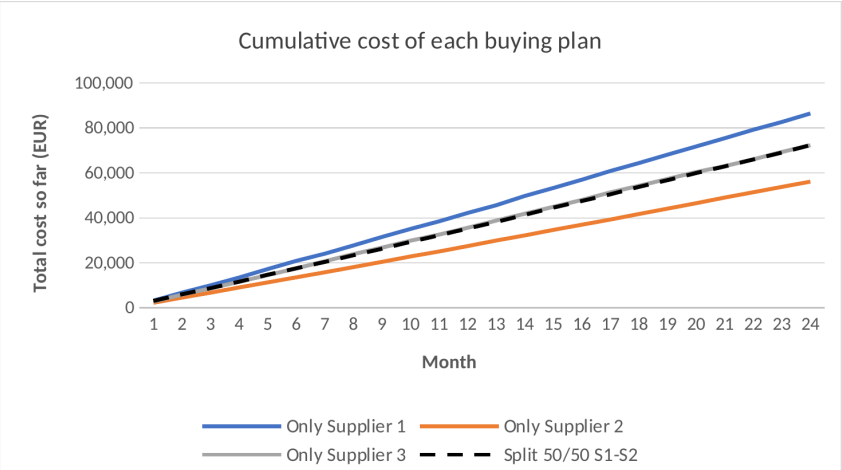


Figure 6.4: Cumulative cost of each buying plan over 24 months

Source: Excel chart of workbook sheet T7 (Table 6.5 shows every third month)

6.6 Tipping Points for the Company

The demo turns the simulation into clear numbers for the managers (expected values; Excel sheet T7):

- Supplier 1 becomes as cheap as Supplier 2 over two years only if one bad rod costs the company more than about **€5,250**, for example through scrap, a customer complaint or a recall. Below this, its better quality does not pay for its price.

Table 6.6: Two-year outcome of each buying plan (mean of 400 simulations)

Plan	Buying (€)	Bad rods (€)	Stops (€)	Total (€)	Bad rods (no.)	Stop h
Only S1	65,679	2,224	18,522	86,426	14.8	61.7
Only S2	43,183	3,145	9,792	56,120	21.0	32.6
Only S3	44,351	4,510	23,460	72,321	30.1	78.2
Split 50/50 S1, S2	55,016	2,670	14,676	72,361	17.8	48.9

Source: demo, 2-year outlook tab: 400 simulations per plan (Excel sheet T7)

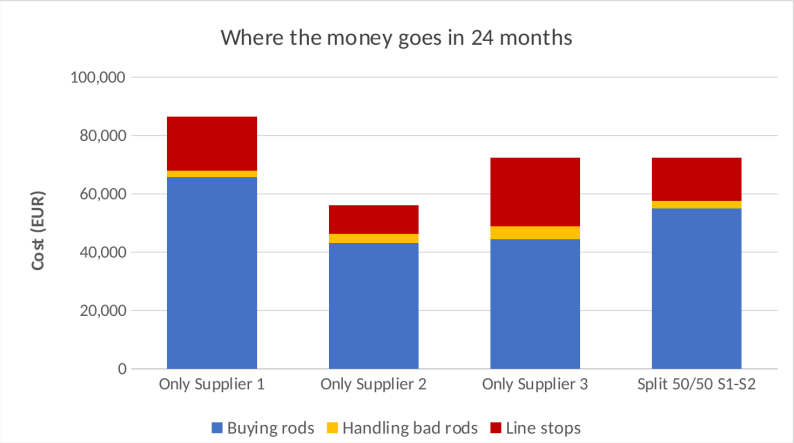


Figure 6.5: Where the money goes in 24 months for each plan

Source: Excel chart of Table 6.6 (workbook sheet T7)

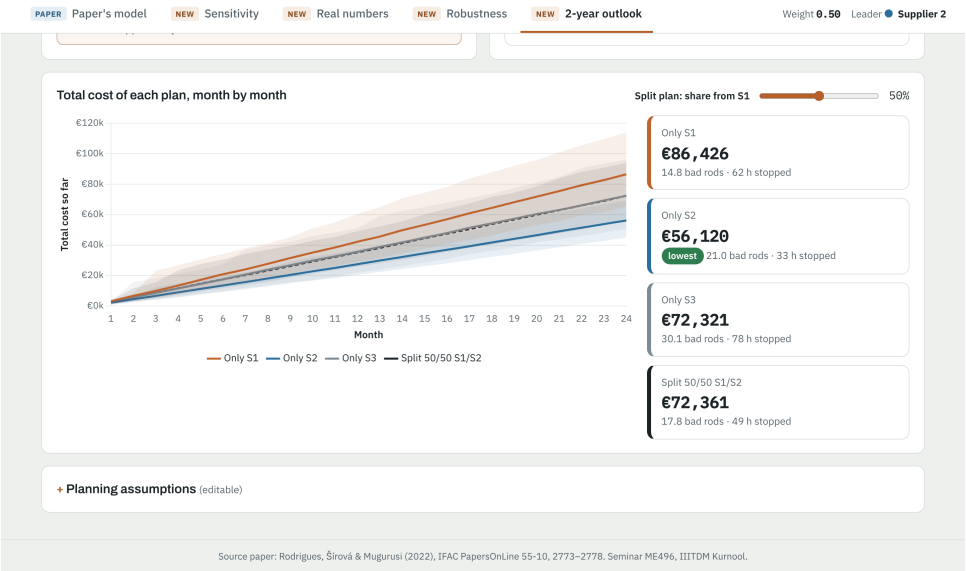


Figure 6.6: Developed demo, 2-year outlook tab: cost lines with 10th–90th percentile band and plan totals

Source: screenshot of the developed demo (own work)

- To match Supplier 2 on total cost, Supplier 1 would have to cut its price from €350 to about **€188** per rod, a 46% discount.
- Supplier 1 causes about **29 more hours** of line stop over two years than Supplier 2, because each wait for a rod lasts 7 days instead of 3.

6.7 Discussion

The paper recommends Supplier 1 because ANP chose it. This work shows that the ANP result comes from one weighting choice, that the two leading suppliers are close under realistic uncertainty, and that Supplier 3 can safely be dropped. When the decision is this close, cost should decide: Supplier 2 is clearly cheaper over two years, and Supplier 1 is justified only if a defective rod is very costly. A split order keeps a second source and lowers the number of bad rods at a moderate extra cost. The cost figures rest on editable estimates, so the company can repeat the analysis with its own data.

Chapter 7

Conclusion and Future Work

7.1 Conclusion

This seminar reviewed a published AHP/ANP model for choosing a copper rod supplier and rebuilt it from its own tables. The AHP result was reproduced exactly. The ANP result was also reproduced exactly as an AHP calculation with the two criteria weighted $1 : 5$, which shows that the change of winner comes from one judgment and not from the network method. The decision flips at a Logistic-Economic weight of 0.442, four of the eight judgment matrices exceed Saaty's 5% limit, and measured quantities were distorted by the verbal scale.

The improved model, built as an offline demo, uses real numbers where they exist, measures confidence with 5,000 random trials, maps the winner for any mix of cost, quality and speed, and simulates two years of cost. Supplier 3 never wins, Suppliers 1 and 2 are nearly tied, and Supplier 2 is cheaper over two years unless a bad rod costs more than about €5,250. A decision model should report not only who wins, but how sure that answer is and what it will cost.

7.2 Limitations of This Work

- The paper gives only ranges for price, lead time and defect rate, so the values for each supplier are estimates inside those ranges.

- The individual judgments of each manager are not published, so their weights could not be studied.
- The cost simulation uses planning estimates for line-stop cost, urgent needs and stock levels. These should be replaced by company data.
- The ANP supermatrix of the paper is not published, so the ANP model could only be matched, not rebuilt step by step.

7.3 Future Work

- Use **fuzzy AHP** so that each judgment is a range instead of a single number.
- Give managers **different weights** based on experience and role, using objective methods such as those reviewed by Koksalmis and Kabak (2019).
- Replace the estimates with the company's **real yearly data** on prices, defects and line stops.
- Find the **best order split** between suppliers with optimisation, following Alegoz and Yapicioglu (2019).
- Connect the demo to **live ERP data**, so that scores and costs update automatically.

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